

DESIGN A SIMPLE CALCULATOR AND SCIENTIFIC CALCULATOR IN R.

Sameer Ahmed G

```
# SIMPLE CALCULATOR IN R
```

```
a <- as.numeric(readline("Enter first number: "))
```

```
b <- as.numeric(readline("Enter second number: "))
```

```
cat("\n1. Addition")
```

```
cat("\n2. Subtraction")
```

```
cat("\n3. Multiplication")
```

```
cat("\n4. Division\n")
```

```
choice <- as.integer(readline("Enter your choice: "))
```

```
if (choice == 1) {
```

```
  result <- a + b
```

```
  cat("Result =", result)
```

```
} else if (choice == 2) {
```

```
  result <- a - b
```

```
  cat("Result =", result)
```

```

} else if (choice == 3) {

  result <- a * b

  cat("Result =", result)

} else if (choice == 4) {

  if (b == 0) {

    cat("Error: Division by zero is not allowed")

  } else {

    result <- a / b

    cat("Result =", result)

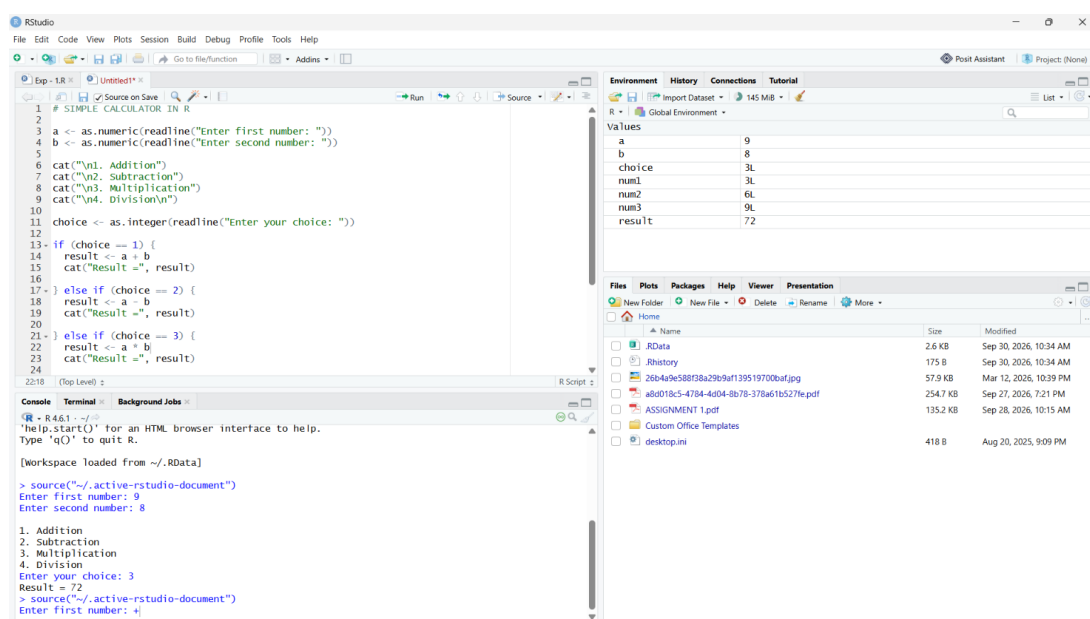
  }

} else {

  cat("Invalid choice")

}

```



```
# SCIENTIFIC CALCULATOR IN R
```

```
x <- as.numeric(readline("Enter a number: "))
```

```
cat("\n--- SCIENTIFIC CALCULATOR ---")
```

```
cat("\n1. Square Root")
```

```
cat("\n2. Power")
```

```
cat("\n3. Logarithm")
```

```
cat("\n4. Sine")
```

```
cat("\n5. Cosine")
```

```
cat("\n6. Tangent")
```

```
cat("\n")
```

```
choice <- as.integer(readline("Enter your choice: "))
```

```
if (choice == 1) {
```

```
  if (x >= 0) {
```

```
    cat("Square Root =", sqrt(x))
```

```
  } else {
```

```
    cat("Error: Square root of negative number")
```

```
  }
```

```
} else if (choice == 2) {
```

```
  y <- as.numeric(readline("Enter the power: "))
```

```
  cat("Result =", x^y)
```

```

} else if (choice == 3) {
  if (x > 0) {
    cat("Logarithm =", log(x))
  } else {
    cat("Error: Logarithm requires a positive number")
  }
} else if (choice == 4) {
  cat("Sine =", sin(x * pi / 180))
} else if (choice == 5) {
  cat("Cosine =", cos(x * pi / 180))
} else if (choice == 6) {
  cat("Tangent =", tan(x * pi / 180))
} else {
  cat("Invalid choice")
}

```

